

## Solution Requirements

|                      |                                          |
|----------------------|------------------------------------------|
| <b>Date</b>          | 27 June 2026                             |
| <b>Team ID</b>       |                                          |
| <b>Project Name</b>  | Rising Waters-ML Flood Prediction System |
| <b>Maximum Marks</b> | 4 Marks                                  |

• **Functional Requirements (FR):** Functional Requirements describe **what the system should do**. They define the core functionalities and services that the flood prediction system must provide to users, such as accepting weather data, processing inputs, generating flood predictions, and displaying the results.

• **Non-Functional Requirements (NFR):** Non-Functional Requirements describe **how the system should perform**. They specify quality attributes such as performance, security, reliability, scalability, and usability to ensure the system operates efficiently and provides a good user experience.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category    | Requirement Description                                                                                                                             | Priority (High/Medium/Low) |
|------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1    | Authentication          | Not Applicable                                                                                                                                      | Not Applicable             |
| 2    | Authorization levels    | Single user access without authentication or role-based authorization.                                                                              | Medium                     |
| 3    | External interfaces     | Accept weather parameters through the Flask web interface and display prediction results to users.                                                  | High                       |
| 4    | Transactions processing | Process user input, apply feature scaling using the saved StandardScaler, execute the trained XGBoost model, and generate flood prediction results. | High                       |
| 5    | Reporting               | Display the prediction outcome (Flood Risk / Safe Condition) along with flood probability and safe probability percentages on the results page.     | High                       |

|   |                                   |                                                                                                                                 |      |
|---|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------|
| 6 | Business rules, etc.              | Use the trained XGBoost model to analyze rainfall and weather parameters and generate predictions based on the trained dataset. | High |
| 7 | Compliance to laws or regulations | Use publicly available datasets and comply with applicable data privacy and ethical AI practices.                               | High |
| 8 | Other                             | Store the trained model (floods.save) and scaler (transform.save) and use them for prediction without retraining each time.     | High |

## Step 2: Non-Functional Requirements (NFR)

| S.No | NFR Category               | Requirement Description                                                                              | Target Metric / Acceptance Criteria                                                                                           |
|------|----------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 1    | Performance & Speed        | The system should generate flood predictions quickly after receiving user input.                     | Prediction should be generated within a few seconds after receiving user input.                                               |
| 2    | Scalability                | The application should support additional weather parameters and larger datasets in future versions. | Support future model and dataset expansion without major redesign                                                             |
| 3    | Security & Data Privacy    | Protect user information and ensure prediction data is securely processed.                           | Secure handling of user input and data                                                                                        |
| 4    | Reliability & Availability | The system should consistently provide accurate predictions and remain available during usage.       | The system should provide consistent predictions and remain available during normal usage.                                    |
| 5    | Usability & Accessibility  | Provide a simple, user-friendly web interface that is easy to navigate.                              | Users should be able to complete a prediction in <b>3 simple steps</b>                                                        |
| 6    | Other                      | Ensure the trained machine learning model maintains high prediction accuracy.                        | Maintain reliable prediction performance based on model evaluation metrics such as accuracy, precision, recall, and F1-score. |